

CSA0609 - DESIGN AND ANALYSIS OF ALGORITHM

Common Elements Between Two Arrays

1. Aim

To write a Python program to find the **common elements between two arrays** using an efficient algorithm and analyze its **time and space complexity**.

2. Algorithm

1. Start with two arrays.
2. Convert the first array into a set.
3. Traverse the second array.
4. Check whether each element exists in the set.
5. If it exists, add it to the common elements.
6. Display the common elements.

3. Pseudocode

Algorithm CommonElements(A, B)

Input: Two arrays A and B

Output: Common elements

1. Create an empty set S
2. For each element x in A do
3. Insert x into S
4. End For
5. Create an empty list Common
6. For each element y in B do
7. If y exists in S then
8. Add y to Common
9. End If
10. End For

11. Return Common

End Algorithm

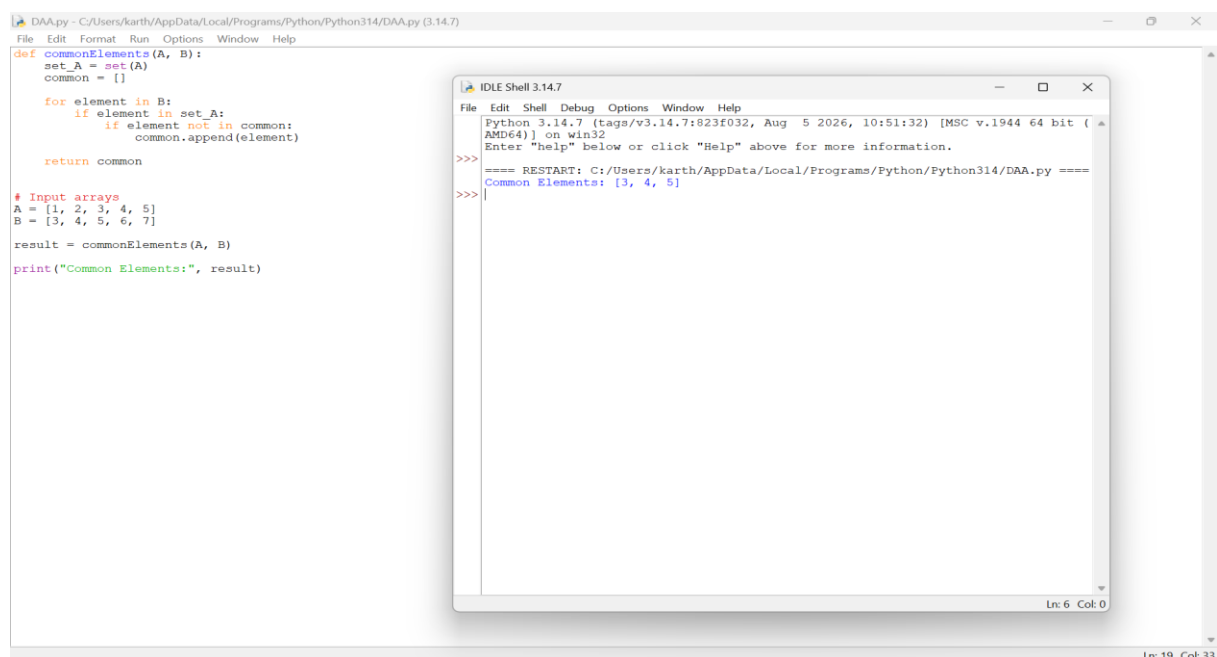
4. Python Code

```
def commonElements(A, B):  
  
    set_A = set(A)  
  
    common = []  
  
    for element in B:  
  
        if element in set_A:  
  
            if element not in common:  
  
                common.append(element)  
  
    return common  
  
# Input arrays  
A = [1, 2, 3, 4, 5]  
B = [3, 4, 5, 6, 7]  
  
result = commonElements(A, B)  
  
print("Common Elements:", result)
```

5. Output

Common Elements: [3, 4, 5]

6. Output Screenshot:



7. Complexity Analysis

Let n be the size of array A and m be the size of array B.

- Creating the set: $O(n)$
- Searching elements of B: $O(m)$ average case

Therefore:

Time Complexity: $O(n + m)$

Space Complexity: $O(n)$

8. Result

Thus, the common elements between two arrays were successfully found using a **set-based approach**, with an average time complexity of $O(n + m)$.